STUDENT HANDOUT

ACM Code of Ethics and Professional Conduct

A practical guide to responsible decision-making in computing

> **“With greater technological power comes greater professional responsibility.”**

1. What is the ACM Code of Ethics?

ACM stands for the **Association for Computing Machinery**. The ACM Code of Ethics and Professional Conduct is a set of principles that guides computing professionals in making responsible, fair, and ethical decisions.

It is relevant to many fields, including:

- Software development and software engineering
- Artificial intelligence (AI) and data science
- Cybersecurity and information systems
- Database management and IT management
- Computing research and digital business

2. Why do we need an ethical code?

Computing technologies can affect privacy, security, employment, financial decisions, education, healthcare, human relationships, democracy, and society. Ethical principles help professionals anticipate risks, reduce harm, protect people, and earn public trust.

Remember: access to technology or data does not mean unlimited permission to use it.

3. The four major sections of the ACM Code

SECTION	MAIN FOCUS
1. General Ethical Principles	Fundamental responsibilities, including human well-being, honesty, fairness, privacy, and confidentiality.
2. Professional Responsibilities	Quality, competence, rules, reviews, risk evaluation, and public awareness.
3. Professional Leadership Principles	How leaders and organizations create ethical workplaces and support responsible computing.
4. Compliance with the Code	Upholding and promoting the principles of the Code in professional practice.

SECTION 1 — GENERAL ETHICAL PRINCIPLES

These principles describe the basic ethical responsibilities of people working in computing.

1.1 Contribute to Society and Human Well-Being

Technology should ultimately benefit people and society.

Example: An AI system helps doctors identify diseases. Before it is used with patients, professionals should assess whether it is sufficiently accurate, safe, and appropriate for that purpose.

1.2 Avoid Harm

Professionals should work to prevent or reduce negative consequences. Harm can include financial loss, privacy violations, discrimination, psychological or physical injury, and security breaches.

Example: A developer discovers a serious security vulnerability but ignores it. Ignoring the problem may expose users to theft or data loss.

1.3 Be Honest and Trustworthy

Professionals should not misrepresent qualifications, hide important information, manipulate data, deceive clients, or make unsupported claims.

Business example: A company advertises that its AI is “99% accurate,” although testing shows only 75% accuracy. This is misleading and unethical.

1.4 Be Fair and Take Action Not to Discriminate

Computing systems should not unfairly discriminate on the basis of characteristics such as gender, age, disability, race, or social background.

AI example: A recruitment algorithm recommends male candidates more often because historical hiring data contains gender bias. The system needs investigation and correction.

1.5 Respect Intellectual Work

Respect other people's original work, including copyright, patents, licenses, and other intellectual property.

Example: A student copies another developer's code and submits it as their own. This is plagiarism and violates intellectual ownership.

1.6 Respect Privacy

Personal information must be collected, accessed, stored, and used responsibly. Examples include names, CNIC or passport details, phone numbers, location, financial and medical information, and browsing behavior.

Discussion question: If a company has access to your data, does that mean it can use it for anything? **No.** Access does not automatically give unlimited ethical permission to use information.

1.7 Honor Confidentiality

Professionals may have access to confidential information and should not disclose it without proper authorization.

Example: A bank employee obtains customers' financial information. Sharing it with friends without authorization is unethical.

SECTION 2 — PROFESSIONAL RESPONSIBILITIES

Computing professionals have duties to the people who rely on their work and to the wider public.

2.1 Strive for High Quality

Produce reliable, secure, usable, and carefully tested work.

2.2 Maintain Professional Competence

Keep learning and updating skills because technology, threats, and standards change rapidly.

2.3 Know and Respect Existing Rules

Understand and follow relevant laws, regulations, policies, and professional standards.

2.4 Accept and Provide Professional Review

Welcome constructive feedback and review the work of others fairly and responsibly.

2.5 Give Comprehensive and Thorough Evaluations

Consider benefits, risks, limitations, and possible consequences before recommending or releasing a technology.

2.6 Improve Public Awareness

Help people understand both the opportunities and risks of computing technologies.

SECTION 3 — PROFESSIONAL LEADERSHIP PRINCIPLES

This section is especially relevant to business students, managers, project leaders, and people who make organizational decisions. Leaders should:

- Support ethical practices and consider public welfare.
- Create appropriate policies and protect users.
- Provide enough time, training, and resources for responsible work.
- Encourage staff to raise concerns without fear of retaliation.
- Avoid pressuring teams to release unsafe or privacy-invasive systems.

Ethical leadership: “We should address the privacy risks before releasing the product.”

Unethical leadership: “Launch it now. We’ll fix the problems later.”

SECTION 4 — COMPLIANCE WITH THE CODE

Computing professionals should uphold and promote the principles of the Code. Ethical responsibility is not limited to avoiding personal misconduct; it also includes taking reasonable action when practices or systems may harm people.

CASE STUDY — THE BIASED RECRUITMENT AI

A company develops an AI system to help recruit employees. After six months, the HR team discovers that the system recommends male candidates more frequently than female candidates. The developer responds:

> **“I didn’t create the bias. The computer made the decision.”**

Questions for students

1. Who is responsible for identifying and addressing the bias?

2. Which ACM ethical principles are relevant to this situation?

3. Should the company continue using the system without changes? Why or why not?

4. What steps should the developer and the company take?

Expected discussion points

- **Avoid harm:** Biased recommendations can unfairly limit employment opportunities.
- **Fairness and non-discrimination:** The company should test for unequal outcomes and investigate their causes.
- **Professional responsibility:** Developers and organizations must not treat an automated decision as an excuse to avoid accountability.
- **Privacy and data responsibility:** Candidate data should be handled lawfully and responsibly.
- **Evaluation and transparency:** The system's performance, limitations, and decision process should be reviewed appropriately.

Recommended actions

- Investigate the training data, model design, and recruitment outcomes.
- Pause or restrict use if the system presents significant unresolved risks.
- Test for bias, document findings, and correct the causes where possible.
- Keep meaningful human oversight in recruitment decisions.
- Inform relevant decision-makers and monitor the system after improvements.

KEY LESSON: Technology does not remove human responsibility.

QUICK REVIEW

Before releasing or recommending a computing system, ask:

- Who could benefit, and who could be harmed?
- Is the system fair, secure, accurate, and respectful of privacy?
- Are the claims about its performance honest and evidence-based?
- Have the risks and limitations been reviewed?
- Who will monitor the system and respond if something goes wrong?

Class takeaway: Ethical computing combines technical competence with honesty, fairness, care for people, and accountability.

Teaching note: This handout summarizes the key ideas supplied for classroom learning. For formal citation or compliance work, consult the official ACM Code of Ethics and Professional Conduct.